Three equivalent descriptions of when a single element is elementary over a generated subsystem

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Editorial note. The counted statement is the author's environment at source0.tex lines 876-883 and its proof environment opens at line 884; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

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Editorial scope. One original result of the pinned author manuscript, counted once: the three equivalent descriptions for one element over a generated subsystem (author label May_26_2024_1).

Allowed standard prerequisites (editorial).

- Closed subsystem. A = <A> means that A is generated by itself, so A is an elementary subsystem of M.. It is the ambient hypothesis.
- Closure of a single element. cl_<(a) = {a} means that the ordering never generates a below itself, which makes a the bottom element above the subsystem.. It is the first of the three conditions.
- Saturated elementary extension. An omega-saturated extension carries realisations of the types that are consistent over the model.. It is where the binary tree of the third condition is realised.

Lemma 0.1 The binary tree of height α has an HF-ordering where the root $a_\emptyset$ is the smallest element instead of being the largest.

Lemma 0.2 Assume $<$ is an HF-ordering of B, $A = \mathrm{cl}_<(A) \subseteq B$ and $C \subseteq B$ is an initial segment such that $c < a$ whenever $c \in C$ and $a \in A$. Then we can interpolate A between C and the rest of the order of B in such a way that we obtain an HF-ordering. In other words,

$$(C, <\restriction C) + (A, <\restriction A) + (B \setminus (A \cup C), <\restriction B \setminus (A \cup C))$$

is an HF-ordering of B. Moreover, we can change the ordering of A by any other HF-ordering of A and still have an HF-ordering of B.

Corollary 0.3 Let $<$ be an HF-ordering of the infinite STS M and assume M is ω-saturated as an $(R, <)$-structure. For every $a \in M$ there is a binary tree $T = \{a_s \mid s \in 2^{<\omega}\}$ in M such that

1. $a_s > a$ for every s
2. $a_s > a_t$ if $s \subsetneq t$.

Proposition 0.4 Assume M is an infinite unconfined STS, $A = \langle A \rangle \subseteq M$ and $a \in M \setminus A$. The following are equivalent:

1. $\mathrm{cl}_<(a) = \{a\}$ for some HF-ordering of M with initial segment A
2. $A \leq Aa \leq M$
3. There are $M' \succeq M$, an HF-ordering $<'$ of M' and a binary tree $T = \{a_s \mid s \in 2^{<\omega}\}$ in M' such that $\mathrm{cl}_{<'}(A) = A$, $a = a_\emptyset$, $T >' A$ and $a_s >' a_{s0}, a_{s1}$ for every $s \in 2^{<\omega}$.

Moreover, these conditions imply $a \leq M$.

Proof. $1 \Rightarrow$ (2 and $a \leq M$). Fix an HF-ordering $<$ of M with initial segment A and such that $\mathrm{cl}_<(a) = \{a\}$. Then we can move a to the bottom (showing $a \leq M$) and we can move a to the top of A, making it the first element of $M \setminus A$ (showing $Aa \leq M$).

$2 \Rightarrow 1$. Since $Aa \leq M$ and $A \leq Aa$, there is an HF-ordering of M having an initial segment Aa, where a is the last element of the segment. Since $a \notin A$, $\mathrm{cl}_<(a) = \{a\}$.

$1 \Rightarrow 3$. Let $<$ be an HF-ordering of M with initial segment A such that $\mathrm{cl}_<(a) = \{a\}$. Let $(M', <') \succeq (M, <)$ be ω-saturated. Then $<'$ is an HF-ordering of M', $\mathrm{cl}_{<'}(A) = A$, $a >' A$ and $\mathrm{cl}_{<'}(a) = \{a\}$. By Corollary 0.3, choose a tree $T_0 = \{a_s \mid s \in 2^{<\omega}\}$ in M' such that $a_s >' a$ for every s and such that $a_s >' a_t$ if $s \subsetneq t$. Then $a \cdot a_\emptyset >' a_\emptyset$, since otherwise the blocks $\{a_\emptyset, a_0, a_1\} \neq \{a_\emptyset, a, a \cdot a_\emptyset\}$ contradict that $<'$ is an HF-ordering. Let $b_0 = a \cdot a_\emptyset$ and $B_0 = \{b_0\} \cup T_0$. Note that $\mathrm{cl}_{<'}(B_0) = B_0 \cup \{a\}$. Now we modify the order $<'$, thus obtaining a new HF-ordering $<_0$ such that $<_0 \restriction B_0 = <' \restriction B_0$ and $<_0 \restriction (M \setminus B_0) = <' \restriction (M \setminus B_0)$.

This ordering $<_0$ is obtained by moving B_0 below a but above A, in particular by declaring that $a >_0 b_0$ and $B_0 >_0 A$ and, hence, $a >_0 a_s$ for every s. This can be done by Lemma 0.2. We have obtained an incomplete tree with root a followed by b_0 on the left and the complete tree T_0 on the right. Now b_0 is a leaf and we should complete the tree below b_0. The construction must be iterated ω times in order to obtain a full binary tree.

$3 \Rightarrow 1$. Let M', $<'$ and T be as in the assumption. Notice that $\mathrm{cl}_{<'}(T) = T$. By Lemma 0.1 T has an HF-ordering $<''$ where $a_\emptyset$ is the smallest element. By Lemma 0.2 we can assume that $<''=<'\restriction T$ and A is an initial segment. Let $<=<'\restriction M$. Then $<$ is an HF-ordering of M, A is an initial segment and $\mathrm{cl}_<(a) = \{a\}$. $\square$